

Sprint Report – Sprint 4

Sprint Goal

Extend the ML pipeline towards production readiness by implementing modelling approaches, experiment tracking, deployment components, and pipeline orchestration.

Sprint Duration

23 March – 29 March 2026

Team Roles

- Faezeh & Sepideh: Modelling, deployment, experimentation, pipeline extension
 - Adi: Scrum master
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Sprint Backlog & Progress

Completed Tasks

- [x] Task 22 — Implement Data Versioning
 - [x] Task 23 — Orchestrate ML Pipeline Using Airflow
 - [x] Task 24 — Dockerize ML Inference Service
 - [x] Task 25 — Research
 - [x] Task 26 — Add unsupervised baseline model
 - [x] Task 27 — Add classical ML baseline
 - [x] Task 28 — Add deep learning baseline
 - [x] Task 29 — Add MLflow
 - [x] Task 30 — API and Deployment Compliance
 - [x] Create data understanding notebooks and documents
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Key Deliverables

1. Modelling Layer

- Implementation of multiple modelling approaches:
 - Unsupervised learning baseline
 - Classical ML models (e.g., SVM, KNN)
 - Deep learning baseline
 - Comparison of model performance:
 - Random split → higher accuracy (~optimistic)
 - Group-based split → poor generalization
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2. Experiment Tracking (MLflow)

- Experiment tracking

- Model versioning
- Metric logging

3. Data Versioning

- Version control for datasets and processed data
- Ensures reproducibility

4. Pipeline Orchestration (Airflow)

- Automated pipeline execution
- Clear workflow structure

5. Deployment & Inference Service

- FastAPI /predict endpoint
- Dockerized service
- Portable deployment

6. API & Deployment Compliance

- Standardized API
- Consistent deployment behavior

Key Insights

- Generalization problem across runs
- Data leakage identified (random split)
- Run variability > gas differences

➡■ Need better features + more robust models

Sprint Metrics

Cycle Time (Estimated)

Task Group	Duration
Modelling	3–4 days
MLflow	1–2 days
Airflow	1–2 days
Deployment	1–2 days
Documentation	1–2 days

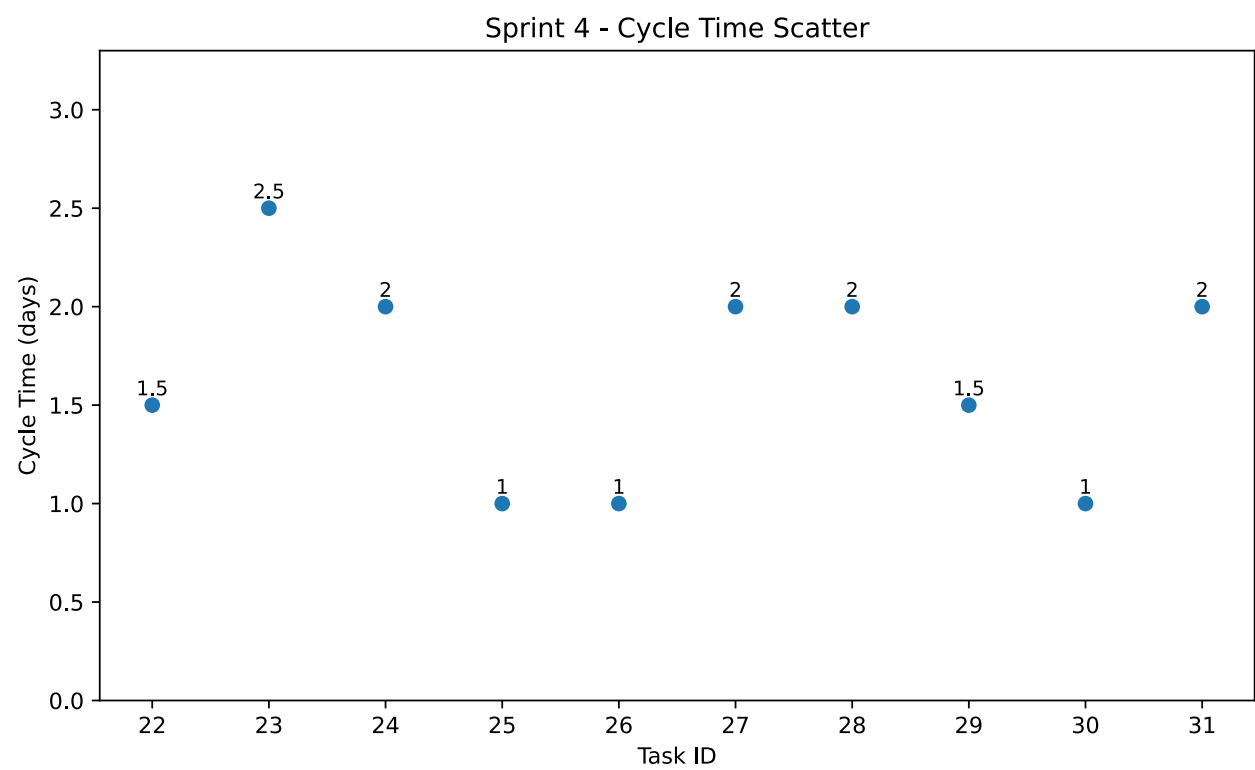
Average Cycle Time: ~2 days

Total Tasks Completed: 10

Completion Rate: 100%

Cycle Time Analysis

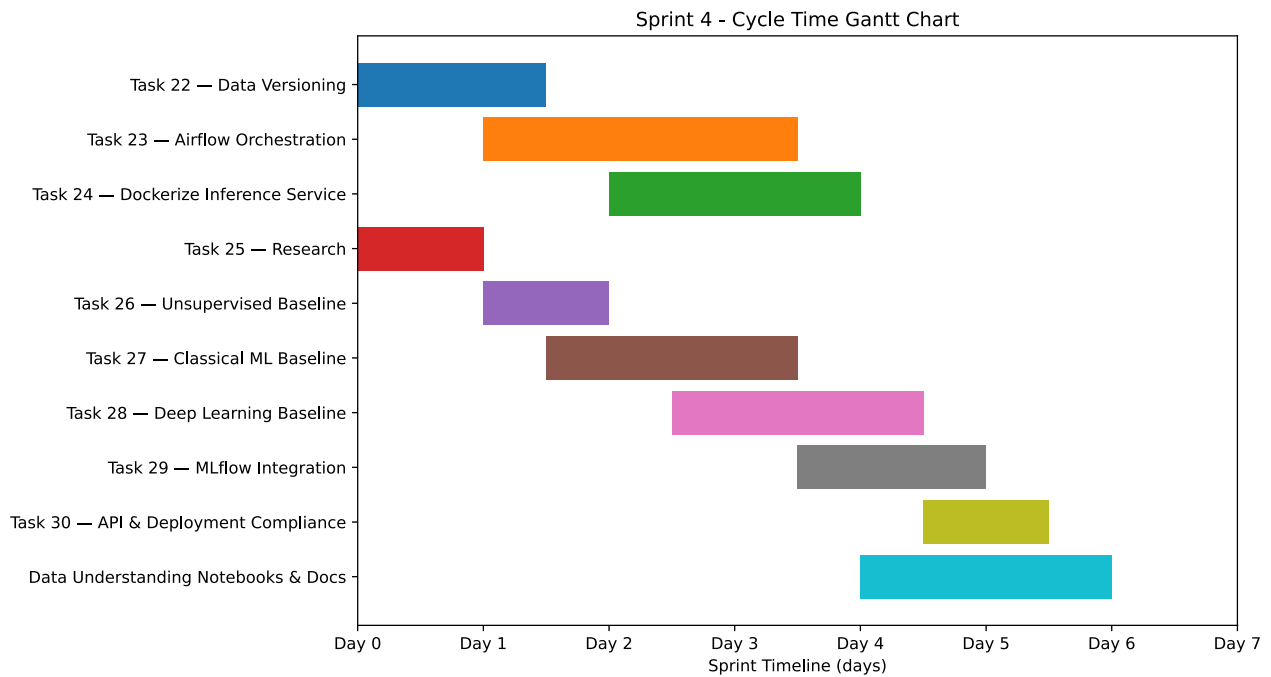
Cycle Time Scatter Plot



Cycle time scatter plot

This chart shows the distribution of task durations. Most tasks were completed within 1–3 days, indicating stable execution despite increased complexity.

Cycle Time Gantt Chart



Cycle time Gantt

The Gantt chart illustrates the progression of work during the sprint, showing a shift from modelling to deployment and integration tasks.

Sprint Summary

Sprint 4 moved the system from a data pipeline to a production-ready ML pipeline.

Key achievements:

- Full modelling stack
- Experiment tracking (MLflow)
- Orchestration (Airflow)
- Deployment (Docker + API)

Main challenge:

- Lack of generalization across runs